Robin Macharg

Senior / Principal iOS & Multi-Platform Software Engineer

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LinkedIn | GitHub | Stack Overflow

United Kingdom | Previous SC1 Security Clearance

Professional Summary

Senior iOS Specialist *focusing on high-performance native Swift and Objective-C application development, with extensive experience shipping secure, consumer-facing apps to the App Store using modern architectures.*

Mobile Security & Encryption *experience highlighted by building custom VPN-secured enterprise clients for the UK Government, integrating MDM solutions, undergoing penetration testing, and holding previous SC1 Security Clearance.*

Scale & Performance Engineering *proven by maintaining core mobile SDKs at Sky serving over 25 million daily active users, optimizing app start-up times, and implementing low-level crash diagnostics for global enterprises.*

Digital Finance & IoT *exposure including developing modern SwiftUI features for Floww's private investment market platform and working with low-level network communications, raw sockets, and embedded platforms.*

Secure iOS & Performance Highlights

- **Government-Grade Security:** Engineered a custom VPN-secured Exchange client for the UK Government's Foxhound project at Amiosec, reverse-engineering SSL traffic and implementing secure MVVM architectures.
- **Enterprise Compliance & Penetration Testing:** Delivered Dyson's first greenfield B2B iPad app in 8 weeks, managing secure enterprise MDM deployment (AirWatch) and successfully passing rigorous penetration testing.
- **High-Scale SDK Performance:** Maintained core video SDKs and reference apps at Sky supporting over 25 million daily active users, prioritizing optimization, analytics, and low-latency delivery.
- **Fintech App Store Delivery:** Developed Composable Architecture (TCA) features on SwiftUI for Floww's App Store-released private investment platform, introducing performance and compile-time monitoring.
- **Low-Level & C++ Crash Analysis:** Analyzed complex stack traces and low-level crash diagnostics across macOS, iOS, watchOS, and tvOS at Bugsnag, ensuring exceptional stability for major enterprise clients.

Current & Recent Experience

Senior Developer | Floww Ltd.

May 2023 – September 2023

Keywords: [iOS, SwiftUI, TCA, Redux, GitHub Actions, CI/CD, Mentoring, Shell Scripting]

- Developed features for a secure private investment market platform using SwiftUI and Redux-like Composable Architecture (TCA).
- Promoted engineering best practices: componentization, API debugging, app startup optimization, and compile-time performance monitoring.
- Maintained robust automation CI/CD pipelines using GitHub Actions and custom automation shell scripting.
- Mentored junior team members and conducted technical code reviews to maintain high quality and security standards.

Lead Mobile Developer | This One Ltd.

May 2022 – May 2023

Keywords: [SwiftUI, Combine, Async/Await, Python, Flask, Fastlane, GCP, Firebase, ML]

- Led greenfield-to-seed-round iOS development as founding mobile developer using SwiftUI, Combine, and async/await architecture.
- Owned full deployment lifecycle, managing Apple Developer accounts, TestFlight distributions, and automated Fastlane pipelines.
- Created a locally deployable mock Python Flask server to decouple backend dependencies and accelerate offline API testing.
- Contributed to core operations including product discovery, UX feasibility, and data modeling for machine learning integrations.

Senior iOS Developer | Sky

October 2021 – April 2022

Keywords: [Swift, Kotlin Multiplatform, SwiftUI, Video Streaming, Analytics, Agile]

- Served as Senior Developer on the Core Video Mobile SDK team supporting apps used daily by over 25 million customers worldwide.
- Maintained hybrid Swift and Kotlin Multiplatform Mobile (KMM) SDKs under tight performance and optimization constraints.
- Rewrote the internal SDK reference app in SwiftUI to showcase modern UI architectures and clean coding patterns.
- Managed bi-weekly release cycles, automated SDK integrations, and implemented repository auto-assign hooks to accelerate Agile workflows.

Senior Developer | Living Map Ltd.

June 2020 – September 2021

Keywords: [iOS, Mapping SDK, XCFrameworks, CocoaPods, Carthage, Background Location, UX]

- Senior developer for flagship iOS Mapping SDKs, overseeing deployment to launch customer Singapore Airlines.
- Re-architected the SDK to support secure background location updates, offline caching, and modern XCFramework packaging.
- Overhauled legacy deployment pipelines, reducing manual release overhead from 3 hours to a single 10-minute automated command.
- Provided technical liaison to client engineering teams to ensure seamless integration into multi-component host apps.

Senior Apple Engineer | Bugsnag Inc.

January 2020 – June 2020

Keywords: [Objective-C, C++, iOS, macOS, watchOS, tvOS, Open Source, Crash Reporting]

- Maintained open-source crash reporting libraries in Objective-C, C++, and Swift for multi-platform Apple ecosystems.
- Provided 3rd-line support handling complex crash log analysis, stack traces, and low-level root cause isolation for enterprise clients.
- Managed high-scale dependency packaging and distribution across CocoaPods, Carthage, Swift Package Manager (SPM), and manual installs.
- Ensured continuous reliability and performance through automated testing pipelines using Travis and Buildkite.

Principal iOS Developer | Amiosec Ltd. (Contract)

March 2019 – December 2019

Keywords: [Xamarin, Visual Studio, C#, Objective-C/Swift Interop, EWS, ReactiveUI, Security]

- Engineered a custom VPN-secured Exchange client for the UK Government's Foxhound project under stringent security protocols.
- Built seamless, secure interoperability between Xamarin/C# and native Objective-C/Swift codebases via .NET Embedding.
- Reversed-engineered secure Exchange Web Services (EWS) using Charles Proxy SSL MiM decryption for API verification.
- Architected reactive MVVM patterns using ReactiveUI and Realm.NET to manage complex offline states and secure local storage.

Principal iOS Developer | Dyson Ltd. (Contract)

June 2017 – January 2020

Keywords: [iOS, Greenfield, Enterprise MDM, ARKit, RealmDB, Swift, Unit Testing]

- Sole developer on Dyson's first B2B iPad app ('Hero'), delivering a production-ready build in 8 weeks with >80% unit/UI test coverage.
- Guided the app through enterprise-grade security penetration testing and MDM deployment using AirWatch.
- Scaled and maintained the application to 23k lines of Swift code, supporting hundreds of worldwide B2B sales representatives.
- Integrated multithreaded RealmDB, custom Xcode templates, and interactive experiences leveraging ARKit and PromiseKit.

Earlier Experience (Summary View)

Head of Mobile Development | Actual Experience Ltd.

March 2014 – June 2017

Keywords: [iOS, Objective-C, Swift, CoreData, Realm, Jenkins, Vagrant, Networking]

- Led mobile development for a greenfield B2B QoS network analytics dashboard.
- Handled ObjC-to-Swift refactoring, CoreData/Realm persistence, low-level ICMP ECHO/TCP networking, and Jenkins CI setup.

Principal Mobile Developer | Accident Exchange Group (Contract)

May 2013 – March 2014

Keywords: [iOS, CoreData, CoreLocation, MapKit, REST, RestKit]

- Developed greenfield consumer iOS app for vehicle telemetry, incorporating CoreLocation and MapKit.
- Implemented CoreData, custom protocols, RestKit integration, and Apple account deployment.

Senior Middleware & DevOps Developer | MySpace

March 2012 – May 2013

Keywords: [Java, Python, Middleware, DevOps, JBoss, Puppet, Scrum]

- Managed Java/Python middleware development and DevOps infrastructure utilizing JBoss, Puppet, and RPM packaging.
- Developed Objective-C second-screen IPTV features and achieved Scrum Master certification.

Earlier Engineering Roles

1996 – 2012

Keywords: [Python, PostgreSQL, OpenGL, PyQt, Linux, Embedded, Java, Perl, Web]

- **Senior Developer | RMS Services Ltd.** (Apr 2009 – Mar 2012): Engineered agentless asset discovery using cross-platform RESTful Python and PostgreSQL.
- **Aerodynamics & Mass Properties Engineer | Airbus UK Ltd.** (Feb 2005 – Feb 2009): Developed 2D/3D wing design and structural analysis tools using Python/PyQt, Perl, and OpenGL.
- **Server & Embedded Developer | Sift / Appliance Studio / Obtree / Circle.com / Glaxo Wellcome** (1996 – 2004): Developed embedded Java/Perl appliances, search infrastructure, and web technologies on Linux.

Open Source & Key Projects

- **Coding Exercises:** Interview algorithms & solutions (GitHub).
- **React Native Input Spinner:** Open-source contributor (GitHub).
- **3D Gear Generator:** Custom 2D and spherical gear design tool built using Vue, TypeScript, and Sass.
- **Bricksmith:** SourceForge award-winning LEGO CAD software for macOS using Objective-C and C (Link).
- **Pycel / Excel Formula Parser:** Python-based Excel dependency graph and formula parser (GitHub).
- **RMHelpfulLabel & Photoshop Scripts:** Custom iOS UI components and batch Photoshop export scripts.
- **Thingiverse Designs:** Parametric 3D CAD models and physical design prints (Designs).

Education, Certifications & Governance

- **BSc (Hons) Artificial Intelligence and Computer Science (Joint)** — University of Edinburgh
- **Certified Scrum Master (CSM)**
- **Parent Governor** — Wyedean Academy and Sixth Form Centre (2016–2020, 2021–Present): Advising on IT infrastructure, Applied Learning, and school financing.
- **Charity Coordinator** — Chepstow Gold Senior Touch Rugby